

Safe Semantic Planner — User Manual

1. Requirements

A C++ compiler is required. The project was developed and tested with GCC through MSYS2 UCRT64 on Windows.

2. Compile

Open the MSYS2 UCRT64 terminal and move to the project folder:

```
cd /c/SafeSemanticPlanner
```

Compile with:

```
g++ -std=c++17 main.cpp LPAStar.cpp Safety.cpp -o planner.exe
```

3. Run

```
./planner.exe
```

4. Entering a Problem

Enter the number of states, followed by each state ID and its coordinates. Then enter the initial state, goal state, bad states, and transitions.

Transition format:

```
ID FROM TO COST SAFETY RELIABILITY AVAILABLE
```

Use 1 for an available transition and 0 for an unavailable transition.

5. Understanding the Output

State Path: sequence of states selected by the planner.

Transition Path: transition IDs used between those states.

Total Cost: sum of selected transition costs.

Minimum Safety Distance: minimum calculated distance to bad states.

Bad States Visited: number of bad states in the final path; a safe solution should normally be 0.

Planning Time: measured planning time in milliseconds.

Estimated Memory Usage: approximate memory used by the planner's main data structures.

6. Example

A problem with initial state 1, goal state 4, and bad state 3 may produce a safe route such as $1 \rightarrow 4$ if the direct transition is available and cheaper or safer than alternatives.

7. Troubleshooting

If the executable does not run, make sure you are using the MSYS2 UCRT64 terminal and that the program was compiled successfully. If a transition is unavailable, enter 0 in the AVAILABLE field. If no safe route exists, the planner reports that no safe path was found.

8. Submission Note

The command-line version is the current submission baseline. A web frontend can be added later without changing the core planner concept.